



Engine Certificate (Quality Assurance)

LR Office
Hamburg
Manufacturer
MTU Friedrichshafen GmbH
Address of manufacturer
MTU Friedrichshafen GmbH
D-88040 Friedrichshafen
Germany

Manufacturer's order number
1196054 / 3145306
Purchaser
Lürssen-Kröger Werft GmbH & Co KG
Purchaser's order number
385420
Intended for ship/yard number
Lürssen Kröger Werft / NB13759
Known engine use
☐ Main ☒ Auxiliary ☐ Emergency

Engine Particulars

Engine serial number
528106062
Type Approval No. and Date
LR2002664TA 02 April 2020
Design Appraisal Document No. and date
ENS 23549-09 02 April 2020
Engine Control
Electronically controlled
Engine Type
☐ 2 Stroke ☒ 4 Stroke ☐ In-Line ☒ Vee ☐ Other:
Shaft power kW and rpm (maximum continuous rating)
3015 kW at 1800 rpm
Number of cylinders
20
Stroke (mm)
210
Diameter of cylinder (mm)
170
Crankshaft ID (Serial No.) & markings
M394 171
Crankshaft Certificate No.
HAM2152064

Fuel Systems

Injection type
Direct injection
Fuel types (select all that are appropriate)
☐ Marine residual fuel
☐ Marine distillate Fuel
☐ Marine distillate Fuel
☐ Low Flash Liquid and/or Gas Fuel
☒ Other
Details
cSt (Max. kinematic viscosity at 50°C)
(☐DMA / ☐DMB / ☐DMC)
(☐DMX)
Specify fuel:
Specify fuel: **DIN 590**

Remarks

Serial nos. of ECU: **258536887** EMU: **257536266** EIM: **257532490** - Only multi plant engine

Engine Identification Marks (QAM No, Engine Serial No and Date)



QAM 030 528106062 LLOYD'S TEST HAM2152268 06 MHS 21

Manufacturer's Statement

This is to certify that the engine described above has been constructed and tested with satisfactory results in accordance with the current Rules and Regulations.

☐ A copy of the applicable Design Appraisal Document and Type Approval Certificate are attached.
☒ Applicable Design Appraisal Document has been reviewed.

Signed on behalf of manufacturer

MTU Friedrichshafen GmbH

Signature

Date

01 July 2021

Name in block capitals

Position in company

Mr. Stotz Wolfgang

Quality Manager

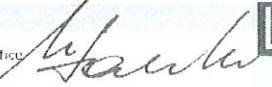
This certificate is issued by the manufacturer in accordance with the arrangements authorised by Lloyd's Register EMEA in Quality Assurance approval Certificate number **QAM 030**

I certify that these arrangements are being kept under review by regular and systematic auditing of the approved Manufacturing and quality control procedures

Date

01 July 2021

Signature

Vienna Office   Lloyd's Register

M.H. Santer
Surveyor to Lloyd's Register EMEA

mhs

A subsidiary of Lloyd's Register Group Limited

This engine will be accepted for fitting in ships classed or intended to be classed with Lloyd's Register subject to satisfactory integration testing of the complete engine with ancillary systems as detailed in Part B of either Form 2011 or 2013 as applicable, and installation under the usual conditions of survey and testing, and where required by the rules, approval of vibration characteristics.

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☐ Main ☒ Auxiliary ☐ Emergency

Engine Particulars

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528106066
Type Approval No. and Date
LR2002664TA 02 April 2020
Design Appraisal Document No. and date
ENS 23549-09 02 April 2020
Engine Control
Electronically controlled
Engine Type
☐ 2 Stroke ☒ 4 Stroke ☐ In-Line ☒ Vee ☐ Other:
Shaft power kW and rpm (maximum continuous rating)
3015 kW at 1800 rpm
Number of cylinders
20
Stroke (mm)
210
Diameter of cylinder (mm)
170
Crankshaft ID (Serial No.) & markings
M650 919
Crankshaft Certificate No.
HAM2152092

Fuel Systems

Injection type
Direct injection
Fuel types (select all that are appropriate)
☐ Marine residual fuel
☐ Marine distillate Fuel
☐ Marine distillate Fuel
☐ Low Flash Liquid and/or Gas Fuel
☒ Other
Details
cSt (Max. kinematic viscosity at 50°C)
(☐ DMA / ☐ DMB / ☐ DMC)
(☐ DMX)
Specify fuel:
Specify fuel: **DIN 590**

Remarks

Serial nos. of ECU: **258536869** EMU: **258539919** EIM: **257532513** - Only multi plant engine

Engine Identification Marks (QAM No, Engine Serial No and Date)



QAM 030 528106066 LLOYD'S TEST HAM2152275 06 MHS 21

Manufacturer's Statement

This is to certify that the engine described above has been constructed and tested with satisfactory results in accordance with the current Rules and Regulations.

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☒ Applicable Design Appraisal Document has been reviewed.

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MTU Friedrichshafen GmbH

Signature

Date

01 July 2021

Name in block capitals

Position in company

Mr. Stotz Wolfgang

Quality Manager

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